



Project Report

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Subject Name: IOT

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Aim: IoT-Based Smart Home Monitoring System using NodeMCU and Blynk

Objectives:

- To interface the NodeMCU (ESP8266) microcontroller with DHT11 and Flame sensors for real-time data collection.
- To transmit sensor readings over Wi-Fi to the Blynk Cloud and display them on a web or mobile dashboard.
- To implement remote manual control of an LED or relay output through the Blynk application.
- To study the performance and responsiveness of the system under various environmental conditions.
- To demonstrate the advantages of IoT in simplifying home and environmental monitoring tasks.

Components Required:

Sno	Name of Component	Qty.
1.	NodeMCU (ESP8266)	1
2.	DHT11 Sensor Module	1
3.	IR Flame Sensor module	1
4.	LED(5mm) + 220 Ω Resistor	1

5.	5V 1A/2A Adapter	1
6.	Breadboard MB-102	1
7.	Jumper Wires	

Details of Components:

1. NodeMCU:

The NodeMCU board is a low-cost, open-source IoT platform widely recognized for its user-friendly approach to building Internet-connected projects. Based on the ESP8266 WiFi chip, it combines a microcontroller with WiFi capabilities, enabling developers to create connected devices with ease

2. DHT11 Sensor Module:

The DHT11 is a digital temperature and humidity sensor. It provides calibrated output signals and is easy to interface with microcontrollers. In this system, it measures environmental temperature and humidity and transmits the data to the NodeMCU.

3. IR Flame Sensor Module:

The IR Flame Sensor detects the presence of fire by sensing infrared light from a flame. It gives a digital signal when a flame is detected. In this project, it helps in identifying fire presence and sends alerts to the NodeMCU and Blynk dashboard.

4. LED (5mm) + 220Ω Resistor:

The LED acts as a visual indicator that lights up when a flame is detected or when activated via Blynk. The 220Ω resistor limits the current to protect the LED from damage.

5. 5V 1A / 2A Power Adapter:

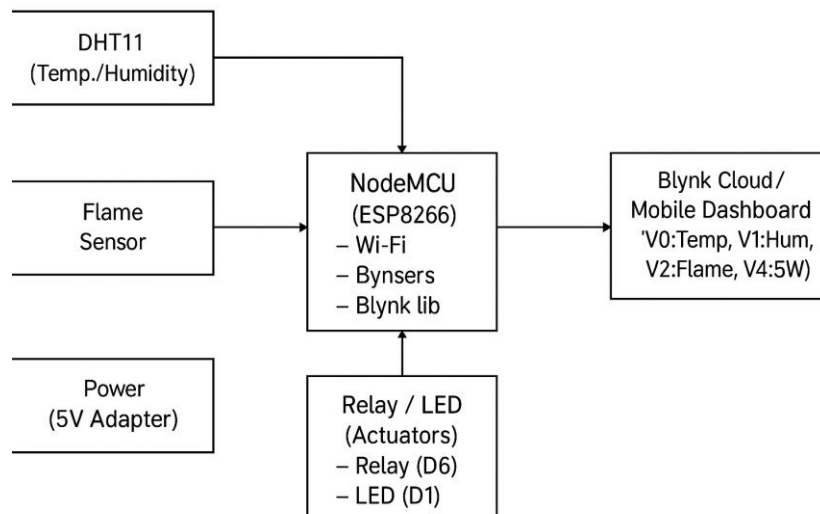
The adapter provides a stable 5V DC supply to power the NodeMCU and sensors. It ensures reliable operation of all components during continuous monitoring.

6. Breadboard (MB-102):

The breadboard is a solderless platform used to connect electronic components for prototyping. It allows easy wiring of the NodeMCU, sensors, and LED using jumper wires.

Block Diagram of Designed Model:

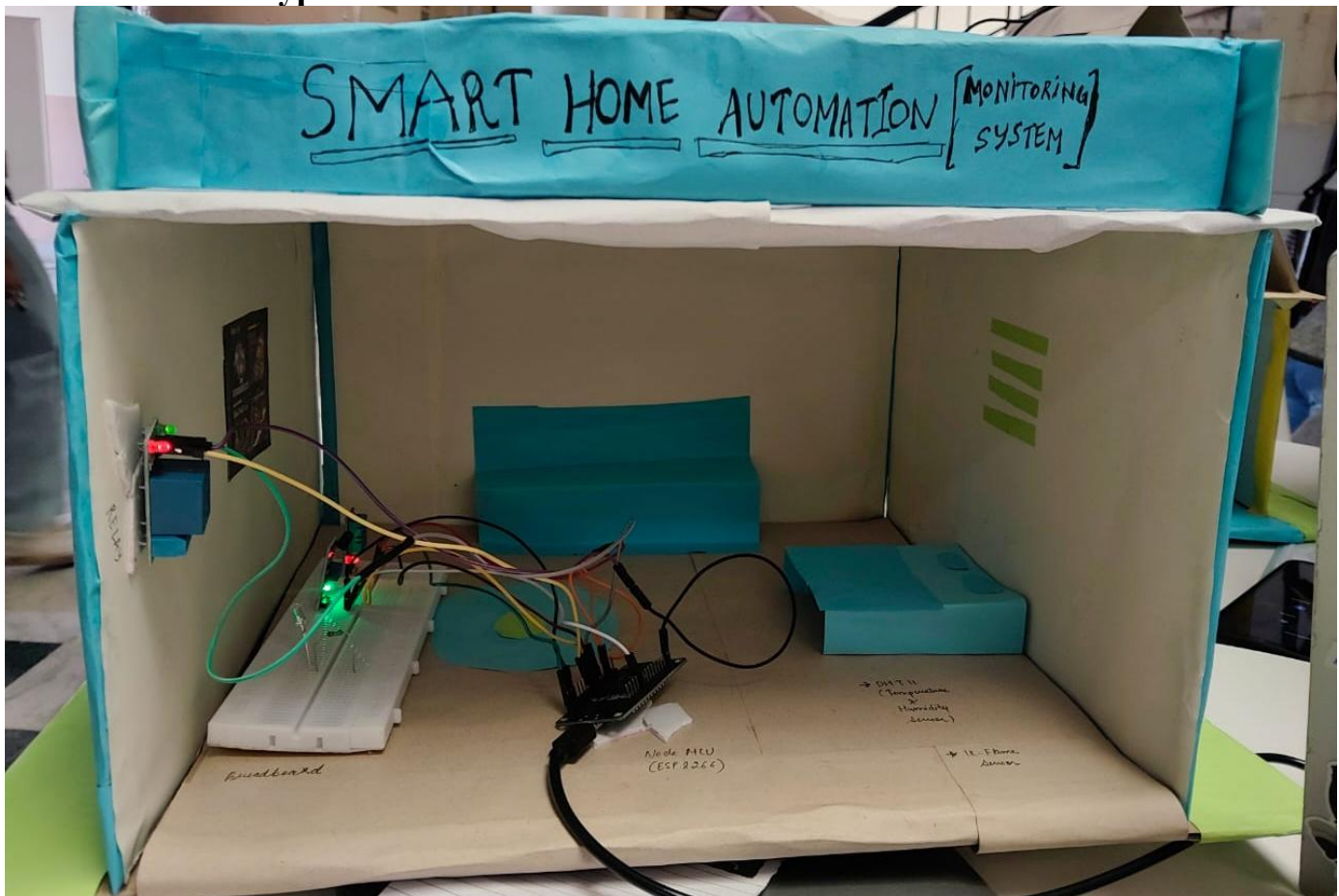
- DHT11 (Temp/Humidity): measures ambient temperature and relative humidity; data read by NodeMCU and sent to Blynk (virtual pins V0 for temperature, V1 for humidity).
- IR Flame Sensor: digital flame presence detector; output read by NodeMCU and mapped to V2 in Blynk (1 = flame detected, 0 = safe).
- NodeMCU (ESP8266): central controller. Reads sensors, runs logic, controls outputs (LED and relay), and connects to Blynk.Cloud over Wi-Fi to update dashboards and receive remote commands.
- Relay & LED (Actuators): Relay switches an external load (fan/light) and LED gives local visual alert. Both are controlled by NodeMCU GPIOs and can be toggled manually from Blynk (V4) or automatically when flame is detected.
- Power (5V Adapter): supplies stable power to NodeMCU and sensor modules (ensure common ground).
- Blynk Cloud / Mobile Dashboard: remote visualization and control. Widgets map to virtual pins:
 - V0 — Temperature display
 - V1 — Humidity display
 - V2 — Flame status (LED widget)
 - V4 — Manual Switch (controls LED/relay)



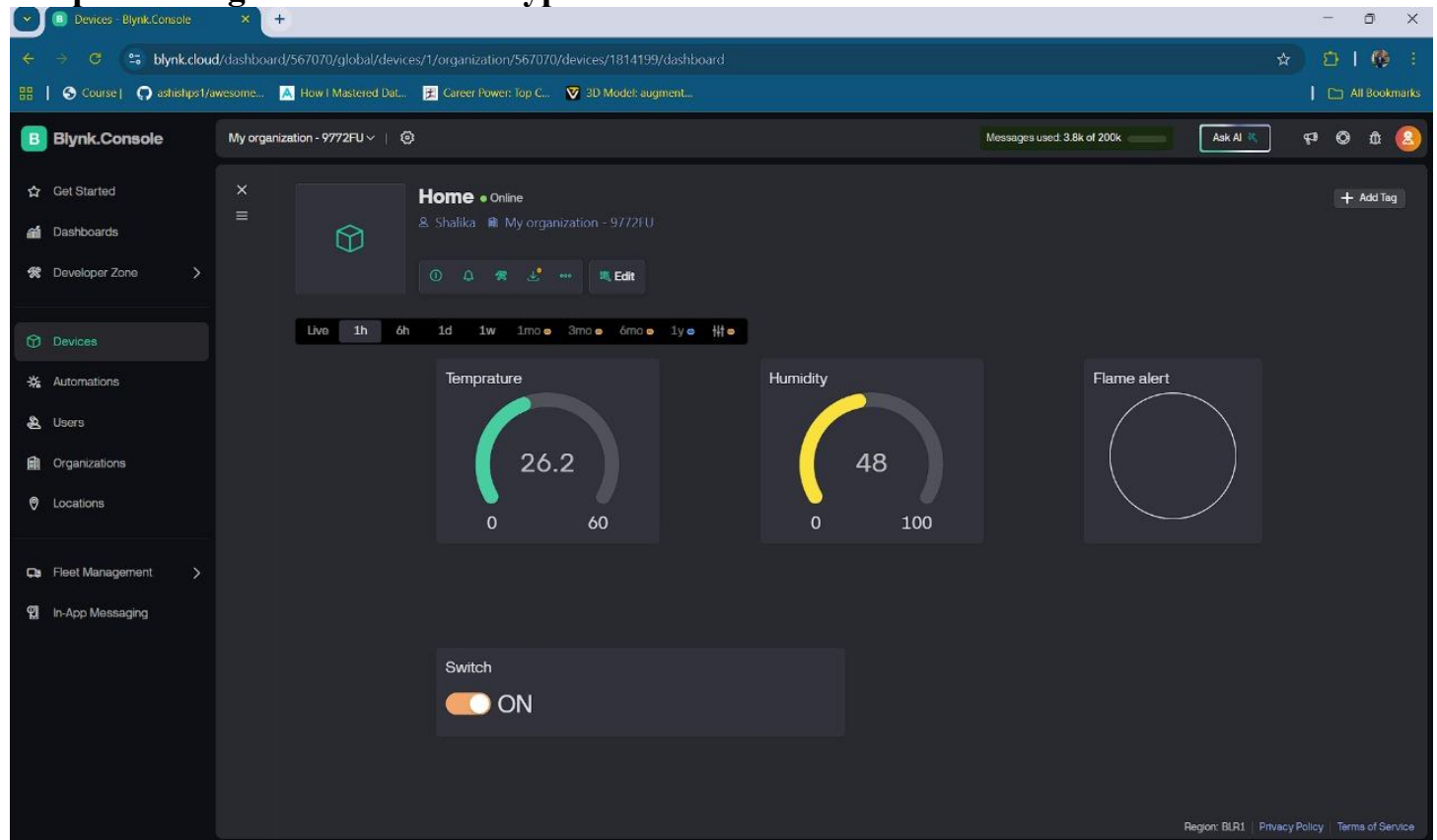
Working of Designed Model:

- The Smart Home Monitoring System is designed to continuously measure and monitor key environmental parameters such as temperature, humidity, and flame presence. The system uses a combination of sensors, microcontroller, and cloud connectivity to collect, process, and display data in real time.
- When the system is powered ON, the NodeMCU (ESP8266) initializes and connects to the local Wi-Fi network using preconfigured credentials. Once connected, it establishes communication with the Blynk Cloud platform using a unique authentication token. The NodeMCU then begins to read data from the connected sensors at regular intervals.
- The DHT11 sensor module measures the ambient temperature and humidity. These readings are converted into digital signals and sent to the NodeMCU, which transmits the values to the Blynk Cloud through virtual pins (V0 for temperature and V1 for humidity). The data is displayed in real time on the Blynk dashboard, allowing the user to view environmental conditions on a mobile or web application from anywhere.
- The IR Flame Sensor continuously detects infrared light in its surrounding area. If a flame or fire is detected, the sensor sends a digital signal to the NodeMCU. Upon receiving this signal, the NodeMCU performs two actions:
 - It activates an LED indicator connected to GPIO D1, alerting the user locally.
 - It updates the Flame Alert widget on the Blynk dashboard (V2), notifying the user remotely through the cloud interface.
- Additionally, the LED or relay module connected to the NodeMCU can be manually controlled via a Switch widget (V4) on the Blynk dashboard. When the user toggles the switch ON, the LED or relay turns ON; when toggled OFF, it turns OFF. This demonstrates remote control capability through IoT.
- All modules are powered using a regulated 5V adapter, which ensures stable operation of the sensors and Wi-Fi module. Communication between the NodeMCU and Blynk Cloud occurs over the Internet using standard TCP/IP and MQTT protocols, ensuring secure and reliable data transfer.

Pictures of Prototype:



Output of Deigned Model/Prototype



Learning outcomes (What I have learnt):

- Practical understanding of sensor interfacing with IoT microcontrollers.
- Experience in configuring Wi-Fi modules and cloud communication.
- Knowledge of Blynk IoT platform and virtual pin data handling.
- Awareness of real-time monitoring challenges and reliability testing.
- Skill development in embedded C/C++ programming using Arduino IDE.

Figure 1 : Output on BLYNK Cloud